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SAGES surgical data science task force: enhancing surgical innovation, education and quality improvement through data science

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What is Surgical Data Science?

Surgical data science (SDS) is an interdisciplinary field that utilizes advances in data science to improve the quality and safety of interventional patient care [1]. Broadly, SDS is the assimilation of information about patients and their environment to assist with healthcare delivery. This encompasses the entire spectrum of the data pipeline, including capture, organization, analysis and modeling, ultimately implemented at various phases of the patient care pathway (Fig. 1). Unsurprisingly, SDS requires the convergence of skillsets from various fields of computer science, mathematics and engineering with domain-specific expertise in surgical care and research, including quality-improvement, education, innovation, epidemiology, health economics, basic science, and others. The term "big data" is often used pervasively to describe the usage of large stores of data for these purposes, and while this is well-established in many non-surgical domains, scaling these methodologies for surgical care has been limited to date, mainly due to the lack of capture, digitalization, storage, and access of procedural data in a meaningful and comprehensive manner.

According to the Surgical Data Science Initiative (http://www.surgical-data-science.org/), the term "surgical data" is inclusive and includes any part of the patient care pathway [2], from initial presentation to long-term outcomes, including all preoperative data (e.g., demographics, disease-specific data, symptomatology, and diagnostic data), intraoperative data (e.g., video feed from image-guided surgery, vital signs, and kinematics data from robotic surgery platform), and post-operative data (e.g., immediate post-operative complications, oncologic outcomes, patient-reported outcomes, and activity tracking). Surgical data can also include data on patients, caregivers, or the technology used to deliver care, which can be acquired from health records, medical devices, or integrated sensors [1]. These large and heterogeneous repositories of data can subsequently be analyzed using complex nonlinear statistical methodologies (such as machine learning) to train algorithms to accomplish a wealth of tasks—everything from computer vision to predicting patient outcomes [3].

Applications

When you consider the evolution of surgical care, data science continues to play a pivotal role in changing our practice over the last century and clinicians, researchers and

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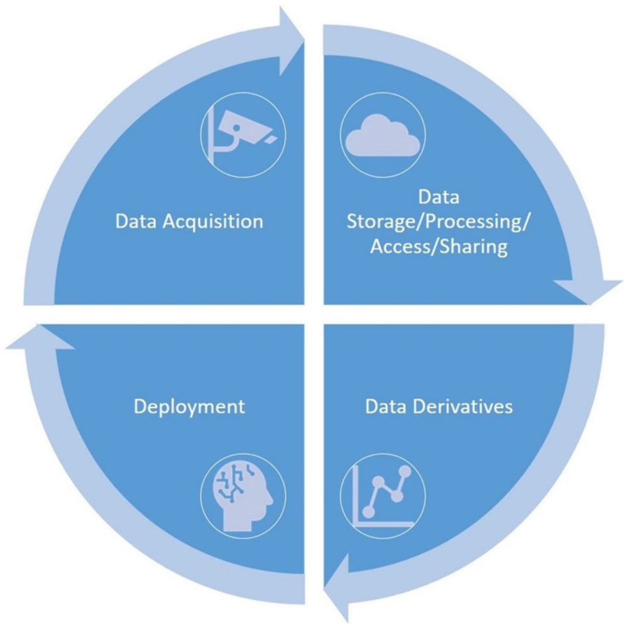

Fig. 1 Data pipeline for surgical data science

institutions are increasingly collecting patient outcomes. Today's clinical environment provides surgical teams with a plethora of data on each patient and it is not uncommon for the operating room to be equipped with advanced imaging, endoscopic platforms and other devices. Nevertheless, it is still up to the surgical team to take this avalanche of data, interpret it using one's domain expertise for each individual patient, and make optimal decisions on a daily basis both in and out of the operating room. If we fast forward to the clinical setting of the future, one can envision an environment where data is seamlessly collected, autonomously analyzed and used to augment and facilitate both perioperative and intraoperative care. This can be in the form of preventive, diagnostic and therapeutic applications throughout the entire continuum of care, as well as quality-improvement for high-level processes and strategies of care or at an individual level for decision support.

One prominent area is intraoperative performance augmentation using computer vision to interpret data from the operating room. For example, numerous algorithms have been developed that are capable of identifying and tracking objects and scenes from the surgical field (e.g. anatomic structures, instruments, phases of the procedure, event detection) [4, 5]. Many of these models have been trained to replicate advanced cognitive behaviors—behaviors that are grounded in expert surgeon mental models that take years of surgical training to acquire [6–8]. While some are designed to provide real-time decision support to the team, other prototypes have also been trained to achieve "context awareness" of the operating room: enabling predictions that are based on environmental and situational data to facilitate processes of care (e.g., improving surgical workflows). This concept has even expanded to tracking operating team members' movements and activities and to understand the operating room as a holistic unit to predict intraoperative events [9]. One area of SDS that may impact patient care in the near future is the use of models that can predict intraoperative events based on anesthesia data, such as vital signs and patient monitoring equipment. For instance, using machine learning, arterial waveform data from patients undergoing non-cardiac elective surgery was shown to predict impending intraoperative hypotensive episodes [10]. It is important to note that most of these studies are proof of concepts and have yet to undergo rigorous validation studies to assess their true performance in the clinical setting as well as their generalizability in real-life scenarios or improvement in patient outcomes to justify their use and adoption. One example is with AI software aimed at improving polyp detection during colonoscopy. While these models have been used enthusiastically and many studies have shown increased polyp detection, the actual evidence for improving detection of neoplasia and patient outcomes is lacking [11, 12].

Finally, another area of impact of SDS that is central to the SAGES mission is in surgical education. Most studies to date in the literature report on the development and validation of algorithms that are able to predict level of expertise, mostly as a summative evaluation (e.g., pass or fail) given input data that ranges from surgical videos, kinematics on robotic platforms, or even eye-tracking data during surgery [13]. Given the "black box" phenomenon of deep neural networks (i.e., algorithms making predictions without actually explaining how those predictions were made), it is unsurprising that there are few examples of models that are able to provide formative feedback to help guide improvement. One notable recent example is by Ma et al., who deconstructed nerve-sparing robotic prostatectomy into granular dissection techniques and were able to correlate which gestures were most predictive of 1-year post-operative erectile dysfunction [14]. Such algorithms will become increasingly common and they will likely be used to provide objective, accurate and reproducible assessment of surgical performance and integrated throughout surgical training curricula, coaching initiatives and for credentialing purposes.

Other areas of SDS that can help improve surgical care include perioperative care, risk prediction, patient flow and resource optimization, automation, and enhancement of robotic surgery. As recent developments over large language models like ChatGPT and Google Bard have shown, the pace of data science and machine learning is accelerating at an unprecedented rate. With newer methodologies and architectures being developed on a regular basis, models are becoming more robust at performing tasks and it is only a matter of time until surgical practice undergoes a digital transformation with SDS as a become a fundamental element. It is, therefore, imperative to establish a strong foundation at SAGES to leverage this technology and innovation to pursue our societal missions in surgical care, education, research and quality.

Obstacles and opportunities for the surgical community

The vision of leveraging SDS to improve surgical care and outcomes at all levels requires overcoming technical, ethical, legal, and cultural hurdles. From a technical perspective, infrastructure is lacking throughout the data pipeline. Piecemeal data collection contributes to a lack of data diversity, the size of video data makes storage costly, the complexity of deep learning algorithms requires even costlier hardware, and metrics to evaluate model performance often lack clinical validity [15]. Annotation of surgical data is time consuming and often requires specific domain expertise in surgical anatomy or procedures, creating a bottleneck in developing models from such data. Both technical infrastructure and theoretical frameworks are also needed to scale the development and translation of any surgical data solutions into clinical practice. While technical problems may not be easy to solve, they are substantially better defined than ethical concerns. The rapid race to solve technical roadblocks in AI development may gloss over issues like personal privacy, model transparency and reproducibility, and biased data [16, 17]. In the context of large-scale surgical data, can individuals still exercise a right to autonomy by requesting their data be removed from datasets? How will surgeons know if SDS algorithms apply to their specific patient populations and if they do apply, who is responsible for outcomes or decisions made using those algorithms? Entities like the United States Food and Drug Administration (FDA) have attempted to set guidelines relating to these questions though the FDA itself recognizes its traditional paradigm of medical device regulation was not designed for current applications of SDS. As such, regulatory guidance is evolving as the FDA and other bodies grapple with how to ensure the safety of SDS-driven decision support software [18].

Beyond regulation, other legal questions remain including data ownership and intellectual property rights when multiple parties in the data pipeline (patients, clinicians, hospital systems, and industry technology providers) may have claims to "own" or access some or all portions of data-driven products. Though the potential is great for SDS to transform surgical practice, these barriers highlight the difficulties as well as opportunities for innovation in leveraging SDS for clinical use.

SAGES and the future of surgical data science

The SAGES SDS task force was created to advocate for and integrate surgical data science within SAGES to advance the society's mission of innovating, educating, and improving patient care. Specifically, the task force is responsible for building a data pipeline and registries of data that will sit on this pipeline, in order to support various applications such as quality-improvement projects, surgical coaching and video-based assessment. Another fundamental pillar of this task force is to set new benchmarks in the field by defining guidelines and best practices (Fig. 2).

The objectives of the SDS task force are to:

1. Procure, implement and maintain an infrastructure for data acquisition and storage to support SAGES initiatives.
2. Develop and maintain a governance strategy for SAGES data registries.
3. Organize academic activities related to Surgical Data Science at SAGES.

To realize the transformational potential of SDS, the surgical community has an opportunity to amalgamate expertise in surgery, education, data science, machine learning, information technology, as well as other disciplines such as ethics, law, global health, and other areas that will undoubtedly intersect with SDS. We have an opportunity to create transcontinental data and machine learning pipelines for creating libraries of algorithms to democratize surgical care and bring expertise to areas that lack access. We have an opportunity to innovate and develop new tools, systems and processes that will enable the global dissemination of scalable surgical artificial intelligence. But more importantly, we have an opportunity to establish the rules of engagement for the safe and ethical introduction of this technology. With a commitment to innovation and improving the quality and safety of surgery, SAGES as an organization is poised to lead this change.

(a)

(b)

Fig. 2 Surgical data science task force engaging in focus groups to **a**) develop a framework that defines various ethical, legal, and other practical considerations for acquiring surgical videos and **b**) define new clinically relevant metrics to measure the performance of computer vision algorithms for minimally invasive surgery (in collaboration with the Surgical Data Science Initiative)

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Declarations

Disclosures Dr. Amin Madani is a consultant for Activ Surgical and Genesis MedTech. Dr. Yao Liu is a research resident at Brown employed by Activ Surgical. Dr. Aurora Pryor is a speaker for Gore, Medtronic and Stryker. Dr. Maria Altieri, Dr. Daniel A Hashimoto, Dr. Liane Feldman have no conflict of interest or financial ties to disclose.

References

1. Maier-Hein L, Eisenmann M, Sarikaya D, März K, Collins T, Malpani A, Fallert J, Feussner H, Giannarou S, Mascagni P, Nakawala H, Park A, Pugh C, Stoyanov D, Vedula SS, Cleary K, Fichtinger G, Forestier G, Gibaud B, Grantcharov T, Hashizume M, Heckmann-Nötzel D, Kenngott HG, Kikinis R, Mündermann L, Navab N, Onogur S, Roß T, Sznitman R, Taylor RH, Tizabi MD, Wagner M, Hager GD, Neumuth T, Padoy N, Collins J,

Gockel I, Goedeke J, Hashimoto DA, Joyeux L, Lam K, Leff DR, Madani A, Marcus HJ, Meireles O, Seitel A, Teber D, Ückert F, Müller-Stich BP, Jannin P, Speidel S (2022) Surgical data science-from concepts toward clinical translation. Med Image Anal 76:102306
2. Maier-Hein L, Vedula SS, Speidel S, Navab N, Kikinis R, Park A, Eisenmann M, Feussner H, Forestier G, Giannarou S, Hashizume M, Katic D, Kenngott H, Kranzfelder M, Malpani A, März K, Neumuth T, Padoy N, Pugh C, Schoch N, Stoyanov D, Taylor R, Wagner M, Hager GD, Jannin P (2017) Surgical data science for next-generation interventions. Nat Biomed Eng 1:691–696
3. Hashimoto DA, Rosman G, Meireles OR (2021) Artificial intelligence in surgery: understanding the role of AI in surgical practice. McGraw Hill Professional, New York
4. Anteby R, Horesh N, Soffer S, Zager Y, Barash Y, Amiel I, Rosin D, Gutman M, Klang E (2021) Deep learning visual analysis in laparoscopic surgery: a systematic review and diagnostic test accuracy meta-analysis. Surg Endosc 35:1521–1533
5. den Boer RB, de Jongh C, Huijbers WTE, Jaspers TJM, Pluim JPW, van Hillegersberg R, Van Eijnatten M, Ruurda JP (2022) Computer-aided anatomy recognition in intrathoracic and -abdominal surgery: a systematic review. Surg Endosc 36:8737–8752
6. Madani A, Vassiliou MC, Watanabe Y, Al-Halabi B, Al-Rowais MS, Deckelbaum DL, Fried GM, Feldman LS (2017) What are the principles that guide behaviors in the operating room?: Creating a framework to define and measure performance. Ann Surg 265:255–267
7. Madani A, Namazi B, Altieri MS, Hashimoto DA, Rivera AM, Pucher PH, Navarrete-Welton A, Sankaranarayanan G, Brunt LM, Okrainec A, Alseidi A (2022) Artificial intelligence for intraoperative guidance: using semantic segmentation to identify surgical anatomy during laparoscopic cholecystectomy. Ann Surg 276:363–369
8. Mascagni P, Vardazaryan A, Alapatt D, Urade T, Emre T, Fiorillo C, Pessaux P, Mutter D, Marescaux J, Costamagna G, Dallemagne B, Padoy N (2022) Artificial intelligence for surgical safety: automatic assessment of the critical view of safety in laparoscopic cholecystectomy using deep learning. Ann Surg 275:955–961
9. Özsoy E, Örnek EP, Eck U, Czempiel T, Tombari F, Navab N (2022) 4D-OR: semantic scene graphs for OR domain modeling. In: Wang L, Qi Dou P, Fletcher T, Speidel S, Li S (eds) Medical Image Computing and Computer Assisted Intervention – MICCAI 2022: 25th International Conference, Singapore, September 18–22, 2022, Proceedings, Part VII. Springer, Cham
10. Wijnberge M, Geerts BF, Hol L, Lemmers N, Mulder MP, Berge P, Schenk J, Terwindt LE, Hollmann MW, Vlaar AP, Veelo DP (2020) Effect of a machine learning-derived early warning system for intraoperative hypotension vs standard care on depth and duration of intraoperative hypotension during elective noncardiac surgery: the hype randomized clinical trial. JAMA 323:1052–1060
11. Wallace MB, Sharma P, Bhandari P, East J, Antonelli G, Lorenzetti R, Vieth M, Speranza I, Spadaccini M, Desai M, Lukens FJ, Babameto G, Batista D, Singh D, Palmer W, Ramirez F, Palmer R, Lunsford T, Ruff K, Bird-Liebermann E, Ciofoaia V, Arndtz S, Cangemi D, Puddick K, Derfus G, Johal AS, Barawi M, Longo L, Moro L, Repici A, Hassan C (2022) Impact of artificial intelligence on miss rate of colorectal neoplasia. Gastroenterology 163:295-304.e5
12. Barua I, Vinsard DG, Jodal HC, Løberg M, Kalager M, Holme Ø, Misawa M, Bretthauer M, Mori Y (2021) Artificial intelligence for polyp detection during colonoscopy: a systematic review and meta-analysis. Endoscopy 53:277–284
13. Lam K, Chen J, Wang Z, Iqbal FM, Darzi A, Lo B, Purkayastha S, Kinross JM (2022) Machine learning for technical skill assessment in surgery: a systematic review. NPJ Digit Med 5:1–16
14. Ma R, Ramaswamy A, Xu J, Trinh L, Kiyasseh D, Chu TN, Wong EY, Lee RS, Rodriguez I, DeMeo G, Desai A, Otiato MX, Roberts SI, Nguyen JH, Laca J, Liu Y, Urbanova K, Wagner C, Anandkumar A, Hu JC, Hung AJ (2022) Surgical gestures as a method to quantify surgical performance and predict patient outcomes. NPJ Digit Med 5:187
15. Reinke A, Tizabi MD, Baumgartner M, Eisenmann M, Heckmann-Nötzel D, Kavur AE, Rädsch T, Sudre CH, Acion L, Antonelli M, Arbel T, Bakas S, Benis A, Blaschko M, Büttner F, Cardoso MJ, Cheplygina V, Chen J, Christodoulou E, Cimini BA, Collins GS, Farahani K, Ferrer L, Galdran A, van Ginneken B, Glocker B, Godau P, Haase R, Hashimoto DA, Hoffman MM, Huisman M, Isensee F, Jannin P, Kahn CE, Kainmueller D, Kainz B, Karargyris A, Karthikesalingam A, Kenngott H, Kleesiek J, Kofler F, Kooi T, Kopp-Schneider A, Kozubek M, Kreshuk A, Kurc T, Landman BA, Litjens G, Madani A, Maier-Hein K, Martel AL, Mattson P, Meijering E, Menze B, Moons KGM, Müller H, Nichyporuk B, Nickel F, Petersen J, Rafelski SM, Rajpoot N, Reyes M, Riegler MA, Rieke N, Saez-Rodriguez J, Sánchez CI, Shetty S, van Smeden M, Summers RM, Taha AA, Tiulpin A, Tsaftaris SA, Van Calster B, Varoquaux G, Wiesenfarth M, Yaniv ZR, Jäger PF, Maier-Hein L (2023) Understanding metric-related pitfalls in image analysis validation. ArXiv
16. Collins JW, Marcus HJ, Ghazi A, Sridhar A, Hashimoto D, Hager G, Arezzo A, Jannin P, Maier-Hein L, Marz K, Valdastri P, Mori K, Elson D, Giannarou S, Slack M, Hares L, Beaulieu Y, Levy J, Laplante G, Ramadorai A, Jarc A, Andrews B, Garcia P, Neemuchwala H, Andrusaite A, Kimpe T, Hawkes D, Kelly JD, Stoyanov D (2022) Ethical implications of AI in robotic surgical training: a Delphi consensus statement. Eur Urol Focus 8:613–622
17. Cobianchi L, Verde JM, Loftus TJ, Piccolo D, Dal Mas F, Mascagni P, Garcia Vazquez A, Ansaloni L, Marseglia GR, Massaro M, Gallix B, Padoy N, Peter A, Kaafarani HM (2022) Artificial intelligence and surgery: ethical dilemmas and open issues. J Am Coll Surg 235:268–275
18. U.S. Food and Drug Administration (2023) Artificial intelligence and machine learning in software as a medical device. https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device. Accessed 05 Aug 2023